

Question number	Answer	Additional guidance	Mark
(a)	An explanation that includes four of the following points: - HIV's host cell is T helper cells / HIV {targets / infects } T helper cells (1) - number of T (helper) cells reduced / T (helper) cells {destroyed / killed} (by HIV) (1) - when HIV enters lytic {cycle / phase} (T helper cells are destroyed) (1) - without T helper cells the (humoral) immune response is {not initiated / weaker} (1) - therefore <i>Mtb</i> are not opsonised (1) - and therefore without antibody phagocytosis (of <i>Mtb</i>) is not so effective (1)	ACCEPT CD4 cells attacks DO NOT ACCEPT other named cells ACCEPT {replication / reproduction} stage comes out of latency ACCEPT B cells not activated person is immunocompromised IGNORE T killer cells not activated ACCEPT no antibodies to bind to <i>Mtb</i> agglutination (TB) bacteria ACCEPT description involving macrophages providing ref to lack of antibodies	(4)

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5(b)(i)	An answer that includes two of the following points: • credit one reason for not testing everyone (1) • credit one reason for not testing everyone (1) • people with HIV may not want to admit they are HIV positive (1) • false negative results / people {only just infected / small viral load} will not test positive (1)	e.g reasons: impossible to test everyone not everyone willing {people don't have symptoms / HIV is dormant } {lack / cost} of {equipment / healthcare} NB if neither mp 1 or mp 2 awarded, 'not everyone is tested' = 1 mark ACCEPT lie ACCEPT false positives	(2)

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5(b)(ii)	• 1×10^7 / 1.002×10^7 (1)	ACCEPT $1 \cdot 10^7$ / $1.002 \cdot 10^7$	(1)

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5(c)(i)	An answer that includes two of the following points: (overall) positive correlation / as percentage of people with HIV increases so does number of people with TB (1) {which is less clear cut / greater spread / weaker correlation} between 0 and {5 / 5.5 / 6 / 6.5} % of HIV (1) - people who have TB do not necessarily have HIV (1)	ACCEPT directly proportional DO NOT ACCEPT causal relationship ACCEPT converse comment for above {5 / 5.5 / 6 / 6.5} % IGNORE some countries have no HIV but do have TB	(2)

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5(c)(ii)	A description that includes three of the following points: - correlation (coefficient) test / named (correlation) test (1) {coefficient / rho / r / r_s / ρ} value calculated (1) - closer (r) value is to {1 / -1} the stronger the correlation (1)	e.g. Pearson, Spearman DO NOT ACCEPT incorrect named test e.g. T test, Z test, MWU, chi squared ACCEPT compare r value to critical values at (less than) {0.05 / 5%} sig levels	(3)

